ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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CO3-10

EXPERIMENT 10

AIM:
To verify the Poynting theorem and to determine the instantaneous and time-average power density (Poynting vector) of a uniform plane wave in free space and in a lossy medium, of a current-carrying conductor, and of solar radiation incident on a panel.

Test case (i) Poynting Vector and Poynting Theorem for a Plane Wave in Free Space

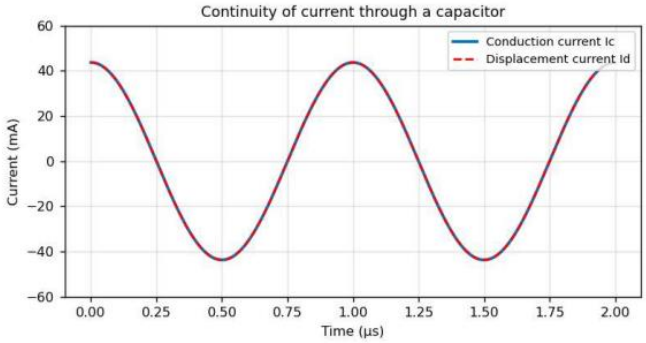
Program:

```
clc;
 clear;
 eps0 = 8.854e-12;
 a = input('Enter plate radius a (m): ');
 d = input('Enter plate separation d (m): ');
 er = input('Enter relative permittivity: ');
V0 = input('Enter peak voltage V0 (V): ');
f = input('Enter frequency f (Hz): ');
 w = 2*pi*f;
 A = pi*a^2;
eps = eps0*er;
C = eps*A/d;
t = linspace(0,2/f,4001);
dt = t(2)-t(1);
 V = V0*sin(w*t);
E = V/d;
% electric field between the plates Jd = eps*gradient(E,dt);
% displacement current density Id = Jd*A;
 % displacement current Ic = C*gradient(V,dt);
 % conduction current in the leads fprintf('Capacitance C = %.4e F\n', C);
fprintf('Peak Jd = %.4f A/m^2\n',
max(Jd(2:end-1)));
 fprintf('Peak Id = %.4f mA\n', 1e3*max(Id(2:end-1)));
 fprintf('Peak Ic = %.4f mA\n', 1e3*max(Ic(2:end-1)));
 plot(t*1e6,1e3*Ic,'b',t*1e6,1e3*Id,'r--','LineWidth',1.5);
```

```
grid on;
 xlabel('Time (\mus)');
 ylabel('Current (mA)');
legend('Conduction current Ic','Displacement current Id');
 title('Continuity of current through a capacitor');
```

OUTPUT:

Sample Output: For a = 0.05 m, d = 1 mm, εr = 1, V0 = 100 V and f = 1 MHz, C = 6.954 × 10^{-11} F, the peak Jd = 5.563 A/m² and the peak Id = Ic = 43.69 mA.



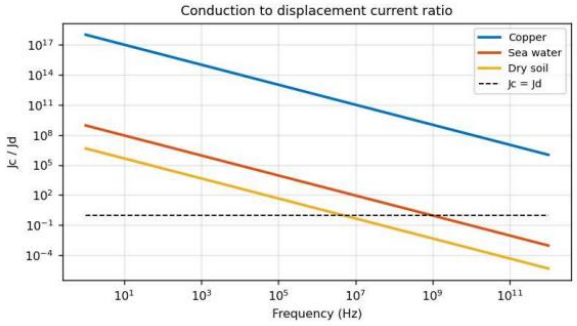
Test case (ii) Conduction and Displacement Current in Different Media
program:

```
clc;
clear;
eps0 = 8.854e-12;
 f = logspace(0,12,500);
 w = 2*pi*f;
 sigma = [5.8e7 4 1e-3];
 er = [1 81 4 ];
 names = {'Copper','Sea water','Dry soil'};
 for k = 1:3 ratio = sigma(k)./(w*eps0*er(k));
% Jc/Jd loglog(f,ratio,'LineWidth',2);
 hold on;
fc = sigma(k)/(2*pi*eps0*er(k));
 fprintf('%s: crossover frequency fc = %.4e Hz\n', names{k}, fc);
end loglog([f(1) f(end)],[1 1],'k--');
```

```
grid on;
 xlabel('Frequency (Hz)');
ylabel('J_c / J_d');
legend(names{1},names{2},names{3},'J_c = J_d');
 title('Conduction to displacement current ratio');
```

OUTPUT:

Sample Output: fc = 1.0426 × 10^{18} Hz (copper), 8.8768 × 10^{8} Hz (sea water) and 4.4939 × 10^{6} Hz (dry soil).



Test case (iii)Numerical Verification of Maxwell's Curl Equations for a Plane Wave:
Program:

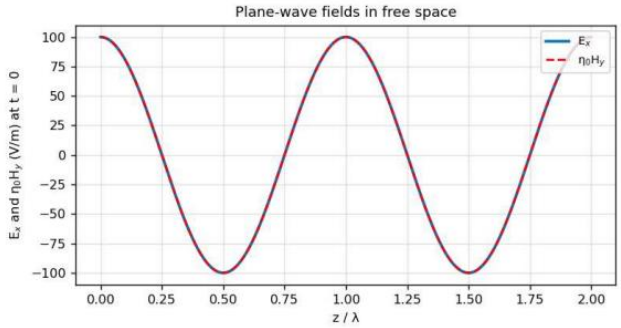
```
clc;
clear;
mu0 = 4*pi*1e-7;
eps0 = 8.854e-12;
 c = 1/sqrt(mu0*eps0);
eta0 = sqrt(mu0/eps0);
E0 = input('Enter peak electric field E0 (V/m): ');
 f = input('Enter frequency f (Hz): ');
w = 2*pi*f;
beta = w/c;
lam = c/f; H0 = E0/eta0;
 z = linspace(0,2*lam,1001);
t = linspace(0,2/f,1001);

 dz = z(2)-z(1); dt = t(2)-t(1);
[Z,T] = meshgrid(z,t);
```

```
Ex = E0*cos(w*T-beta*Z);
Hy = H0*cos(w*T-beta*Z);
 [dEx_dz,dEx_dt] = gradient(Ex,dz,dt);
[dHy_dz,dHy_dt] = gradient(Hy,dz,dt);
res1 = dEx_dz + mu0*dHy_dt;
 % Faraday: dEx/dz = -mu0 dHy/dt res2 = -dHy_dz - eps0*dEx_dt;
 % Ampere-Maxwell (J = 0): -dHy/dz = eps0 dEx/dt idx = 2:1000;
 % interior points err1 = max(max(abs(res1(idx,idx))))/max(max(abs(dEx_dz(idx,idx))))*100;
err2 = max(max(abs(res2(idx,idx))))/max(max(abs(eps0*dEx_dt(idx,idx))))*100;
fprintf('beta = %.4f rad/m, wavelength = %.4f m, H0 = %.4f A/m\n', beta, lam, H0);
fprintf('Residual of Faraday law = %.3e %%\n', err1);
fprintf('Residual of Ampere-Maxwell law = %.3e %%\n', err2);
plot(z/lam,Ex(1,:),'b',z/lam,eta0*Hy(1,:),'r--','LineWidth',1.5);
grid on;
xlabel('z / \lambda');
 ylabel('E_x and \eta_0 H_y (V/m) at t = 0');
legend('E_x','\eta_0 H_y');
 title('Plane-wave fields in free space');
```

OUTPUT:

Sample Output: For E0 = 100 V/m and f = 1 GHz, β = 20.9582 rad/m, λ = 0.2998 m, H0 = 0.2654 A/m and the residuals of both equations are below 10^{-6} %.



Test case (iv) Magnetic Field Produced by Displacement Current (Ampère–Maxwell Law):

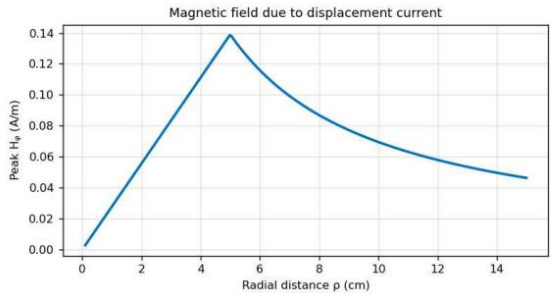
program

```
clc;
 clear;
eps0 = 8.854e-12;
```

```
 a = 0.05;
d = 1e-3;
er = 1;
V0 = 100;
f = 1e6;
w = 2*pi*f;
 eps = eps0*er;
 Jd_m = eps*w*V0/d;
% peak displacement current density Id_m = Jd_m*pi*a^2;
% peak displacement current rho = linspace(0.001,0.15,300);
H = zeros(size(rho));
inside = rho <= a;
 H(inside) = Jd_m*rho(inside)/2;
 % rho <= a H(~inside) = Id_m./(2*pi*rho(~inside));
% rho > a plot(rho*100,H,'LineWidth',2);
 grid on; xlabel('Radial distance \rho (cm)');
 ylabel('Peak H_\phi (A/m)');
title('Magnetic field due to displacement current');
 fprintf('Peak H at rho = a : %.4f A/m\n', Id_m/(2*pi*a));
```

OUTPUT

Sample Output: The peak Hφ at ρ = a is 0.1391 A/m; Hφ = 0.0695 A/m at ρ = 2.5 cm (inside) and 0.0695 A/m at ρ = 10 cm (outside).



Result :

Thus the Verification of Maxwell's Equations and Displacement Current in Time-Varying Electromagnetic Fields using MATLAB successfully.

LABORATORY PERFORMANCE ASSESSMENT RUBRICS

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weighting**	**Marks given**	**Marks obtained**
Identify Maxwell relations and assumptions	Identify Maxwell relations and assumption s	Selects the appropriate integral/differential equation and identifies time varying quantities.	Selection is correct with minor omissions.	. Needs guidance choosing relations.	Uses inappropriate equations.	25	2.5	
Calculate displacement current and field quantities	Calculate displacement current and field quantities	Implements the displacement current relation and associated field calculations correctly.	Calculations are mostly correct with minor errors.	Partial calculations require guidance.	Calculations are incorrect or absent.	25	2.5	
Verify equations numerically or graphically	Verify equations numerically or graphically	Demonstrates consistency of the chosen Maxwell relation with computed values/plots and units.	Verification is mostly correct.	Verification is incomplete.	No valid verification is demonstrated.	30	3	
No valid verification is demonstrated.	Explain physical significance	Explains displacement current and its role in time varying fields with clear documentation.	Explanation is generally correct.	Explanation is superficial or incomplete.	Cannot explain the result.	20	2	
TOTAL						100	10	